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The adoption of automation technology for the mitigation of material waste on construction sites in the Malaysian construction industry

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ABSTRACT

The construction business has always been at the forefront of technological advancements, and this trend has only accelerated in recent years as one of the important contexts among the other numerous new technologies. The current study aims to examine the relationship between Automation Technology Adoptions (ATA) and Construction Site Performance (CSP) and their influence on material waste mitigation measures in Johor construction sites. Data was gathered utilizing an online survey method, with the participants from the Johor construction industries representing contractors and experts. A total of 257 responses were valid to investigate the assumptions of the validations, and 'Partial Lease Square - Structural Equation Modeling' (PLS-SEM) procedure has been used. The adopted model may be used to help construction site managers and workers understand the importance of the ATA, CSP, and their utilization as material waste management approaches. The obtained results emphasize the relevance of automation technology adoption for mitigating construction material waste in construction sites with such a clear significance and clarity under the use of automation in construction sites. These findings will assist construction professionals to manage their materials on the job site and complete their projects.

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Automation technology adoption; material waste mitigation measure; site performance; Malaysian construction industry

Introduction

Automation Technology (AT) in construction has quickly developed globally as a trend and innovation that plays a venture by its tools and devices to produce high-quality work with fantastic nature and efficiency based on its current technological revolution and control in the construction industry (Rangani 2020). A study by Kuppusamy et al. (2019) asserts that, given the emphasis on the cost of energy, the incorporation of green building materials by stakeholders in Johor Bahru's construction industry manifests various attributes beneficial to the green building materials sector. This adoption addresses a paramount concern within the Malaysian construction industry, underscoring the significance of such materials in meeting the industry's critical priorities. These green materials comply with international and Malaysian government requirements. GBI in Malaysia provides a solid foundation for construction by evaluating environmental design and efficiency using six essential components. These include energy conservation, interior environment protection, sustainable site development, management materials, supplies allocated, water efficiency, and innovative thinking (Uche et al. 2013; El-Abidi et al. 2019). Industrialized building systems (IBS) are cutting-edge production techniques that boost construction's effectiveness and reputation. In Malaysia, the term 'IBS' was explicitly established in the early 2000s to incorporate the usage of off-site, prefabricated, and standardized components. The goal of Malaysia's Construction Sector Transformation Plan 2021–2025 aims to double the production of the construction industry by establishing an 'industrialized building system' (IBS) to speed up the construction process, make the site safer for workers, result in higher-quality construction, and to reduce project costs (Thomas Tarang et al. 2022). Using (IBS) as a contemporary method of construction has many benefits, such as shorter construction times, reduced site materials placing, effective site management, minimal waste, a focus on the environment with clean sights, lower labors intensity, construction consistency, and lower overall capital costs (Abedi et al. 2011). The construction sector must consider IoT network size increases; otherwise, it will fall behind other industries. This is a consequence of the fact that the construction sector should employ IoT applications and solutions to ensure the industry's efficient operation (Mahmud et al. 2018). Therefore, this study aims to empirically test and investigate the Automation Technology Adoption utilizing GBI, IBS, and IoT as material waste mitigation measures to enhance Construction Site Performance (CSP). Material Management Adoption (MMA) moderates the association for this quantitative approach to investigate its influence as a moderator between ATA and CSP

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practices in the Malaysian construction industry. This research is primarily concerned with material waste mitigation measures through the use of automation technology. Construction site performance is fueled by the growing concern of man's activities on the construction site. Furthermore, the Material Management Adoption (MMA) is adopted as a moderator due to the similar functions in material waste management to find out its influence and interaction in either strengthening or decreasing the relationship between ATA and performance. In addition, a previous study has found that the lack of efficient and proper waste management technologies has increased construction costs and resource waste (Wong and Roslan 2019). Malaysia has experienced significant advancements and transformations in its construction project development sector over the years. Positioned strategically in the southern region of Malaysia, Johor Bahru has witnessed substantial growth, largely attributed to its advantageous proximity to Singapore (Yap et al. 2019). The establishment of the Iskandar Malaysia development corridor in 2006 has proven to be a pivotal catalyst, greatly influencing construction and real estate development in this region (Rahim et al. 2017). The construction sector in Johor Bahru remains dynamic, shaped by both governmental policies and active private sector participation. This dynamic environment renders Johor Bahru a noteworthy subject for scholarly investigation within the domain of urban development and construction in Southeast Asia. It is crucial to note that Johor is a state located in the southern region of Malaysia. The knowledge gap addressed in this research is to investigate the association of Automation Technology adoption as material waste mitigation measures toward construction site enhancement. Since there has been no extensive study into automation technology utilizing GBI, IBS, and IoT as one direction for CSP. The adoption of automation technology at construction sites could cover resource management throughout the entirety of the project's lifecycle, beginning with the early stages of planning and design and continuing through construction as recommended by Hoeft et al. (2022).

Research framework and hypothesis development

The green building idea is relatively new to Malaysia and requires innovative materials, it is challenging for developers, contractors, and subcontractors to obtain all of the necessary materials and technologies for the construction of green construction (Chin et al. 2020). It has been shown that green practices and techniques in the construction sector significantly benefit the environment. Green building has been characterized as the current trend in environmental protection. However, various advantages and obstacles are drawing and restraining the developers from using and implementing GBI in their projects. There are substantial differences in values produced from the green construction index, which comprise reducing, reusing, and recycling material waste in construction sites (Oyenuga and Bhamidimarri 2015). This concept appeared to optimize energy conservation, conservation of natural resources, and waste reduction. Considered to be the most conventional method for lowering the amount of construction waste that is diverted to landfills, the value of wastes discovered by decreasing material waste includes strategies, recognition programs, sufficient capital and support for the implementation. However, greener efforts must be recalibrated in terms of new technologies, the supply of essential information, management, and guidelines.

H_1: There is a significant relationship between Automation Technology Adoptions of 'Green Building Index' (GBI) and Construction Site Performance (CSP).

An industrialized building system IBS is a construction method that makes use of tools, materials, or structural components to be installed as prefabricated parts at the worksite. Furthermore, Affendi et al. (2017) stated that IBS implementation is significant for reducing noise by providing ready elements to be installed in the working place. Through careful design and execution, prefabricated elements might decrease the quantity of waste on the worksite (Rofdzi and Ismail 2018). According to the benefits of IBS use for construction waste reduction and its economical usage that lowers the total construction costs stated by Pozin et al. (2017), industrialized building systems (IBS) practice is gaining increasing attention for a construction site, particularly as it reduced the dependency on foreign labors, and site labor numbers as well. The finding may hypothesize that IBS technology and its innovation in construction sites might be one of the most efficient ways to improve the performance of a building site.

H_2: There is a significant relationship between Automation Technology Adoptions of 'Industrial Building Systems' (IBS) and Construction Site Performance(CSP).

The Internet of Things (IoT) for construction can capture certain types of data on such activities, actual progress, and conditions on the site and then transfer this information to a centralized panel, where the data is assisted in the formation of choices about the construction process (Lin et al. 2019; Sidani et al. 2021). Because of this, the Internet of Things not only makes use of the most recent technology in the software sector, but it has become more affordable over the decades to employ the technology adoptions as innovations for the construction industry.

H_3: There is a significant relationship between Automation Technology Adoptions of 'Internet of Things' IoT and Construction Site Performance (CSP).

The communication between all parties at the construction site is one of the material management indicators that provide sufficient information between all parties. It was shown to impact building site performance significantly (Oseghale et al. 2021). Effective collaboration and communication between all stakeholders are essential, given the nature of the work (Setiawan et al. 2021). Besides, Abdul Razak et al. (2017), Crawford et al. (2017), and Gamil et al. (2020) mentioned that value engineering is one of the material management aspects with higher specifications and less cost for construction site management.

The strategic management, storage, transportation, and distribution of materials used in the construction sector include efficient inventory management, which plays a crucial part in the practical completion of the task since its primary goal is to guarantee that the resources are accessible at the point of usage when required (Ashika 2019). According to those definitions, materials management is commonly understood as selecting, coordinating, and regulating the proper material source of the precise quality at the appropriate time and location for the lowest-cost building process (Kulkarni et al. 2017).

H_4: There is a significant relationship between 'Material Management Adoptions' (MMA) and Construction Site Performance (CSP).

The GBI uses a similar methodological framework to other tools, such as energy and efficiency (Subki and Mahazir 2019). As new technologies of urban area expansion, the innovation of

GBI practice has been presented in Malaysia, particularly the green innovation tools that have been accepted by the local government to promote the concept of green construction or technology adopted as a means of changing in the constructing market to reconfiguring the prevalent a way of thinking about setup and start to operate to reduce the carbon emissions. Green construction materials demonstrate a symbiotic relationship between the environment, humans, and material resources (Kamil et al. 2018). It is helpful to take the insights from the execution and management of the existing nationwide green building construction to hasten the growth of the construction industry (Zhao and Moh 2016)

H_5: Material Management Adoption (MMA) moderates the relationship between Green Building Index (GBI) and Construction Site Performance (CSP).

Materials management in construction recently adopted various technologies in the firm; IBS technology is a modern advanced technology (Pozin et al. 2017). The Malaysian construction industry enjoys the advantages of producing higher structures, on-time construction finishing, and cost savings through the uniformity of IBS. On the other side, development is employed to design the overarching strategy for the construction industry (Azira et al. 2020). As a result, the developer needs to work closely to address their efforts throughout the implementation stage to attain a better efficiency level. However, evaluating the amount of cooperation in construction companies has proven to be challenging.

Furthermore, performance must be measured systematically to assist the relevant parties in determining the contractor's competence and commitment to complete the duties given. The evaluation will also assist in identifying areas that need improvement and measure how the participants feel about the finished image or outcome. In sustainable buildings, actual environmental performance is crucial, and the inadequate monitoring of ecological performance may impede workers' performance. Environmental obligations, issues, policies, green information systems, and employee assessment should be the common subjects of a green performance evaluation through implementing IBS and its benefits in material and cost saving. These areas should cover job-related requirements and volunteer external and internal environmental initiatives (Brinly 2020).

H_6: Material Management Adoption (MMA) moderates the relationship between the 'Industrial Building System' (IBS) and Construction Site Performance (CSP).

The ultimate goal of utilizing IoT in construction projects can be seen as the tools and equipment to achieve optimum utilization of available resources currently offered with robust system planning, with controlled expenditures, at the expense of negligible waste and minimal consequences, as IoT can monitor the component and facilitate the equipment's management to remain on track and observed the massive through a variety of data within the entire project lifecycle instantly (Maru and Raval 2020). IoT technique enables enormous promise for boosting on-site hygiene, effectiveness, and efficiency. Companies may increase workplace visibility at all phases of construction activities, including the post-construction processes, in actuality by implementing IoT sensors and gadgets. Companies are utilizing IoT technologies with several potential advantages as the construction industry is modernizing quickly (Edwards 2020). Moreover, the Integrated platform is used for administration at the construction site, which instantly puts an extra order for supplies supply when a certain threshold is reached (Lut and Takki 2019).

H_7: Material Management Adoption (MMA) moderates the relationship between the 'Internet of Things' (IoT) and Construction Site Performance (CSP).

Figure 1 below illustrates the conceptual model guiding this research as mitigation measures as the independent variables, the material management adoptions as the moderator, and the dependent variable is construction site performance.

Research methodology

Research methodology involves the systematic process of collecting, analyzing, and interpreting data to answer research questions and achieve study objectives. One popular approach to constructing a research methodology is the 'research onion' model (depicted in Figure 2), developed by Saunders and Lee (2017) which outlines the main stages involved in formulating an effective methodology. These stages begin with defining the underlying philosophy, selecting approaches, methods, and strategies, and defining time horizons, ultimately leading to the research design - the main techniques and procedures for data collection and analysis as different strategies, practices, or procedures are used by researchers in social science to conduct a research. Saunders and Lee (2017) as they summarized the strategies and stages of conducting research in one visual representation as can be observed in Figure 2. This figure was used in this research to determine the research design.

First, this research follows the positivism approach by Wimalaratne and Kulatunga (2022). This philosophy was preferred because it uses a large sample size with no involvement for the researcher in the problem domain. Besides, there is no subjectivity in the data. All of this reduces the dais in the results. In addition, this philosophy is appropriate because the current research is based on developing hypotheses, testing them, and then generalizing the results on the population. Second, this research follows the deductive approach. This is because the current research does not establish a new theory. However, this research was based on previous theories to establish the hypotheses. In addition, this research followed the survey strategy to collect data. In this study, the populations are contractors and experts in the Malaysian construction industry. This research employed a quantitative methodology, utilizing a questionnaire survey administered to key stakeholders within the construction sector of Johor, a state within Malaysia. The study aimed to investigate the adoption of Automation Technology for Material Waste Mitigation through Material Management practices to enhance Construction Site Performance. This survey investigates respondents' evaluations of mitigation measures in the construction industry, and the use of technologies and material management adoptions for reduction and it's proposed due to the statement above. The respondents in the firm chosen based on their positions as directors, engineers, architects, project managers, quantity surveyors, developers, local authorities/government agencies, and researchers/academic institutions due to the need of this state to enhance construction site performance during the implementation of materials.

A preliminary investigation was carried out to validate the study and establish the problem statement. The survey was administered online and in hard-copy format for this investigation. The researcher collected data from six respondents from different organizations/companies of solid waste management, recycling, and construction management companies in Johor and other states of Malaysia. The exploratory research was useful in validating the necessity for the present study, which concentrates

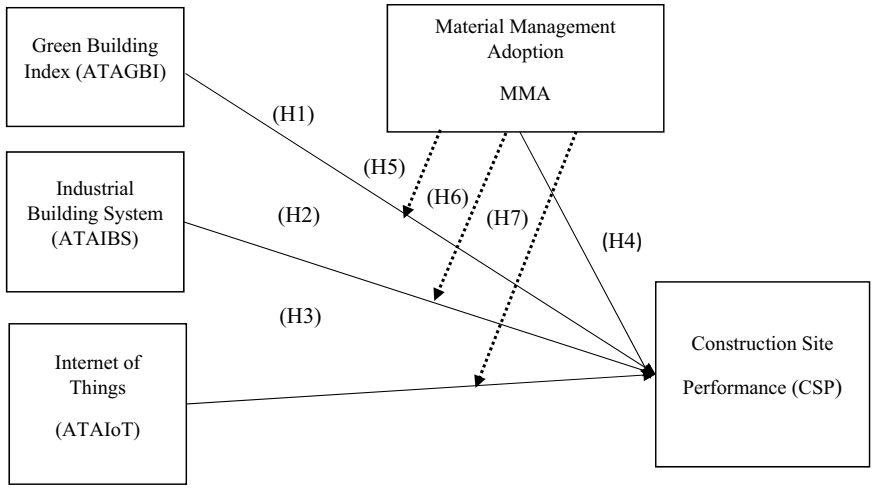


Figure 1. Conceptual framework.

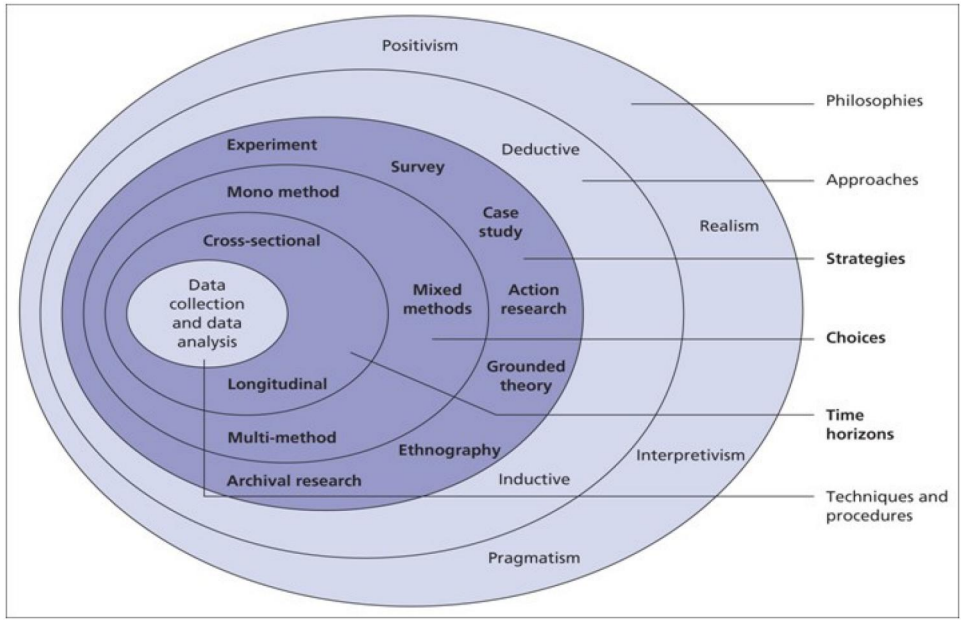


Figure 2. Methodology research onion source: (Wimalaratne and Kulatunga 2022).

on the employment of automation technology adoption utilizing GBI, IBS, and IoT moderated by MMA as material waste mitigation toward construction site performance. The modified questionnaire survey was undertaken with experts to identify any issues with the survey questions. One way to conduct the questionnaire pre-testing is by taking the opinion of professionals from the same field to screen the questionnaire's survey. During the screening process, the professionals are asked to focus on the wording, difficulties, and gathering variables, and notice the major issues they suggested. The procedure administered the pre-test with 5 professional lecturers from the faculties of construction engineering and civil engineering, as well as some actual engineers who had a good part of expertise to handle these concerns from the study area at the Johor construction industry. The feedback gathered from the pre-test is typically helpful in exploring how to alter specific inquiries' language and format.

A pilot study was carried out before the precise data collection. In this study, forty (40) sets of the survey were given to the participants, such as lecturers, academics, and construction parties in the Johor construction industry. The researcher successfully collected a total of 33, of which the valid sets were 30 sets, with three sets containing missing parts. The utilized questionnaire in this study was adapted from previous studies. First, items the 'Green Building Index' (GBI) is an automation technology to enhance environmental measures on the construction site. The measurement items for this construct were adapted from Marhani and Muksain (2018) and Manzoor et al. (2021). The second item of measurement was the adaption of IBS technology as an offsite construction method for environmentally friendly development and its ability to enhance construction performance in the current research were adapted from Algburi et al. (2018), Turkyilmaz et al. (2019), and Azira et al. (2020).

Per previous studies on material management, Ibrahim et al. (2021) and Mallak et al. (2014), developed that the relation of material waste reduction focused more on the time, cost, and quality of construction projects in their investigations. The implementation of material management practices within the construction industry has been recognized as a strategy that enhances project delivery processes and improves overall quality standards. Additionally, it contributes to cost-effectiveness by ensuring that buildings meet required standards, aligning with

prevalent practices in the construction sector (Rotimi 2022). Furthermore, IoT was the fifth construct in this research and its measurement items were adapted from Dallasega (2018), Mao et al. (2019), Perrier et al. (2020), Ibrahim et al. (2021), and Dilakshan et al. (2021). A study by Alaloul et al. (2016) found that in British construction projects, approximately 30% of buildings require rework, resulting in a 40%–60% decrease in labor efficiency, with material waste accounting for 10%. The study, conducted among stakeholders, also mentioned that to reduce the material waste impact in construction sites, there is a motivational concern indicating that innovation and technologies, as suggested by the participants, can guide how to manage the material waste at construction sites effectively. Performance measures rely on having the appropriate people with the right skills and equipment to produce projects on schedule and within budget and having the right materials in the proper location. Income stream and capital are also given careful attention to purchasing the materials and labor (Kulkarni et al. 2017). At the same time, site performance is enhanced positively by estimation, budgeting, planning, and monitoring materials in the site workplace (Chowdhury et al. 2019).

To ensure efficient construction site operations, including equipment and material flows, it's crucial to address timing and spatial conflicts in asset storage and transportation while effectively managing site progression. This involves optimizing construction site layout and resource management (Whitlock et al. 2018) Efficient materials management in the construction sector encompasses strategic storage, transportation, and distribution, with a primary focus on ensuring timely resource availability (Ashika 2019). Materials management is commonly defined as the coordination and regulation of material sourcing in terms of quality, timing, and cost-effectiveness (Kulkarni et al. 2017). A model was developed to explore the boundary conditions and moderating factors influencing the adoption of automation technologies and materials management at the Malaysian construction industry. Knowledge sharing plays a significant role in moderating the relationship between automation technology adoption and site performance, particularly in enhancing construction waste mitigation and overall site efficiency.

In this study, the focus lies on the population comprising contractors and seasoned professionals within the Johor construction firm, including architects, project managers, and academic researchers. This selection aligns with the latest updates and reflects the pertinent stakeholders actively involved in shaping the landscape of the Malaysian construction industry within the Johor region (CIDB, 2019). The maximum number of respondents required for this research is 365 respondents, as determined by Krejcie and Morgan (1970) and granted by Rahi (2017) based on the sample size calculation for a population of 7,481. The questionnaire was administered online using the Google website and emails. A total of 257 valid responses were collected, representing a response rate of 70%, to investigate and assess the assumptions of the validations. The evaluation was a five-point scale, 1 = Strongly Disagree, 2 = Disagree, 3 = Neither Agree nor Disagree, 4 = Agree, 5 = Strongly Agree (Zikmund et al. 2013; Ajayi and Oyedele 2018).

Advanced statistical tools, Statistical Package for Social Sciences SPSS 25.0 and Structural Equation Model/Partial Least Square (PLS-SEM), notably SmartPLS version 3.7.9, were used to achieve dependability in evaluating the obtained data (Sarstedt et al. 2019). The inferential approach was used to assess the conceptual framework study for the hypotheses using (PLS-SEM) to evaluate the characteristics of respondents for obtaining the findings of this study.

Data and results analysis

Respondent's profile

The study involved participants from various roles within the construction industry, including directors, engineers, architects, project managers, quantity surveyors, academicians/researchers, and others. Designer/consultant firms and contractors were the primary sectors represented, with buildings being the most common project type. Project costs ranged from less than 1 million to over 20 million Malaysian Ringgit, with the majority falling in the 1 million RM categories. Most participants held certificates, and the majority had 0–5 years of professional experience as shown in Table 1.

Reliability test

The value of 'Cronbach's Alpha' CA is considered excellent if (>0.9), Good if (0.7, 0.9), Acceptable if the value ranges between (0.6, 0.7), and Poor if the value ranges between (0.5, 0.6), ultimately (0.5) determined as unacceptable (Ajayi and Oyedele 2018). Evaluating the overall survey's variables with more excellent reliability indicates a better consistency for questionnaire fields and the overall mean of the questionnaire. Cronbach's coefficient alpha has a typical range of from 0.0 to 1.0, with higher values representing more consistency. The Cronbach's alpha coefficient

Table 1. Demographic output, frequency, and percentage.

NO	Item	Frequency	Percentage %
1.	Positions		
	Director	78	28.9%
	Engineer	105	38.9%
	Architect	20	7.4%
	Project manager	26	9.6%
	Quantity surveyor	6	2.2%
	Academician/researcher	28	10.4%
	Others	7	2.6%
2.	Organization		
	Designer/consultant firm	88	32.6%
	Contractor	73	27.0%
	Manufacturer	22	8.1%
	Client	14	5.2%
	Developer	21	7.8%
	Local authority/government agency	13	4.8%
	Research/academic institution	31	11.5%
	Others	8	3.0%
3.	Type of project		
	Building	137	50.7%
	Infrastructure	54	20.0%
	Institutional and commercial	42	15.6%
	Industrial	20	7.4%
	Other	17	6.3%
4.	Cost of project		
	< 1 million	123	45.6%
	1–5 million	40	14.8%
	6–10 million	35	13.0%
	6–10 million	28	10.4%
	16–20 million	25	9.3%
	>20 million	19	7.0%
5.	Education's level		
	Certificate	110	40.7%
	Diploma	28	10.4%
	Master degree	61	22.6%
	PhD	71	26.3%
6.	The working experience		
	0–5 years	123	45.6%
	6–10 years	55	20.4%
	11–15 years	48	17.8%
	16–20 years	29	10.7%
	>20 years	15	5.6%

Table 2. The reliability test of research constructs.

Constructs	Cronbach's Alpha	No. of Items
ATAGBI	0.897	4
ATAIBS	0.928	8
ATAIoT	0.928	14
MMA	0.944	11
CSP	0.932	15

Table 3. The descriptive statistics for the latent variables.

Latent construct	Number of items	Mean	Std. deviation
ATAGBI	4	3.68	0.802
ATAIBS	8	3.92	0.679
ATAIoT	14	3.90	0.718
MMA	11	3.82	0.642
CSP	15	3.71	0.718

was calculated within all the survey items, as shown in Table 2 which displays the 'Cronbach's Alpha' coefficient.

Descriptive analysis for the latent constructs

The questionnaire results are summarized by adopting the 5-point Likert scale, whereby 5 points indicate a strong agreement and a scale of 1 indicates strong disagreement (Louis and Dunston 2018). The overall findings of the questionnaire. Table 3, presents the results of the descriptive statistics below.

Descriptive statistics in Table 3, revealed that Automation Technology Adoptions utilizing the Industrial Building System obtained a higher mean value than the remaining four variables (3.92), followed by Automation Technology Adoptions Using the Internet of Things (3.90), Material Management Adoptions (3.82), and Construction Site Performance has a mean value of (3.71). In addition, ATA utilizing GBI has a low mean value of (3.68). These statistics show that the mean scores are higher than the average.

Assessment of the model measurement

Measurement models in research require a comprehensive assessment of item reliability, internal consistency, content validity, discriminant validity, and convergent validity (Hair and Sarstedt 2019). The outer model, assessed using PLS-SEM, serves as a tool to examine the measurement model.

Table 4, shows that the Cronbach alpha for all variables is more than 0.70. As a result, all variables in the present investigation are consistent (Wan and Gtmmad 2013). Furthermore, Table 4 shows that all factors revealed that all variables have strong reliability, and their AVEs are higher than threshold values > 0.5, supporting the measurement model's reliability (Bido and Silva 2019). Thus, overall reliability the latent variables used in this study had internal consistency values ranging from 0.853 to 0.934 for the components of the study. These values are satisfactory since they are all above the minimum threshold of 0.70. Chin et al. (2009) proposed that to achieve adequate CV, the AVE of each variable should be 0.50 or higher, while the threshold for composite dependability is 0.70. Table 4, displays the (AVE) values ranging from 0.662 to 0.765. As a result, it can be concluded that CV has been established for each construct in this research since each item reflects the latent components. As illustrated in Table 5, the significant values of each of the AVE items along the diagonal lines are 'greater than the corresponding values in both columns and rows,' confirming the discriminant validity assumption (Fornell and Larcker 1981).

As a general rule, discriminative validity occurs if the value is lower than the restrictive value of 0.85 of HTMT test (Henseler et al. 2016). As can be seen in Table 6, below, all of the HTMT values are less than the restrictive value of 0.85, indicating that discriminant validity does exist for the HTMT. This means that HTMT criteria detected no collinearity issues among the latent constructs multicollinearity. As a result, there are no intersecting construct elements, and they weren't used to measure the same thing.

Structural model for direct relationships

The structural model is evaluated when the model has been completed (Henseler et al. 2016). Figure 3, presents the output data obtained from determination, R-value, the factor loading, using SmartPLS 3.3.9. The hypothesis judgment for each hypothesis significance level was established based on these measured values. Despite this, the t-values in the present study are the result of bootstrapping (with 5000 sample rounds for 257 instances/observes) Table 7. The bootstrapping in this research is applied to assess the moderating effect as Hair and Sarstedt (2019) recommend that researchers implement a bootstrapping test using the sampling distribution of the indirect impact, which works for simple and complex moderator models.

Assessment of hypothesis testing (path coefficient)

Coefficient of determination (R^2)

Henseler et al. (2016) proposed the criteria for R^2 levels. First, if the values of R^2 are more than 0.67, the result indicates a high level while between 0.33 and 0.67 indicates a moderate level. However, if the value of R^2 obtained ranges between 0.19 and 0.33, then the level is considered small, and if the value of R^2 indicates 0.19 or less, the evaluation is deemed to be unacceptable Nitzl et al. (2016). As a consequence of the findings of this research, all R^2 values have matched Henseler et al. (2016); and Hamdollah and Baghaei (2016) requirements. The R^2 values are presented below in Table 8.

Based on the data presented in Table 8, it was found that R^2 is high in explaining ATAGBI, with a value of 0.675, reflecting 67.5% for the coefficient of determination throughout CSP. In addition, the R^2 of the CSP was moderate with the value of ATAIBS indicated at 0.555% out of the endogenous Construct CSP with a moderate explanation following the R^2 of ATAIoT, MMA accounted for 0.641,0.634, as a part of CSP variance, respectively.

Assessment of effect size (f^2)

As a result, values that is more than 0.35 are rated as significantly high, and if the values range between 0.15 and 0.35 have to be considered medium, while the values between 0.02 and 0.15 are considered small. In addition, if values are less than 0.02 have no impact as guided by Cohen (1988); Hair and Sarstedt 2019).

According to Table 9, the f^2 of ATAGBI, ATAIBS, ATAIoT, and (MMA) effects on CSP were moderate, high, small, and small, respectively, as the finding revealed, and ATAIBS on CSP was medium (Hair and Sarstedt 2019).

Predicative relevance (Q^2)

Table 10, shows both the 'sum of squared observations' (SSO), commonly known as SSO, and the 'sum of squared prediction errors' (SSE), commonly known as SSE. Additionally, Table 10 displays the relevancy of Q^2 (1-SSE/SSO). As a result, this

Table 4. Reliability reflective, loading, average variance extracted of constructs summary.

Constructs	Item	Loadings > 0.70	Cronbach's Alpha	Composite Reliability	AVE > 0.5
Construction site performance	CSP1	Deleted Due to Low Loading	0.932	0.925	0.681
	CSP2	0.817			
	CSP3	0.833			
	CSP4	0.821			
	CSP5	0.849			
	CSP6	0.840			
	CSP7	0.834			
	CSP8	0.871			
	CSP9	Deleted Due to Low Loading			
	CSP10	0.807			
	CSP11	0.823			
	CSP12	0.808			
	CSP13	0.809			
	CSP14	0.817			
	CSP15	0.800			
Green building index	ATAGBI1	0.866	0.897	0.853	0.765
	ATAGBI2	0.889			
	ATAGBI3	0.891			
	ATAGBI4	0.852			
Industrial building system	ATAIBS1	0.825	0.938	0.921	0.699
	ATAIBS2	0.836			
	ATAIBS3	0.828			
	ATAIBS4	0.81			
	ATAIBS5	0.833			
	ATAIBS6	0.859			
	ATAIBS7	0.840			
	ATAIBS8	0.856			
Internet of things	ATAIoT1	0.821	0.928	0.911	0.709
	ATAIoT2	0.819			
	ATAIoT3	0.835			
	ATAIoT4	0.832			
	ATAIoT5	0.845			
	ATAIoT6	0.839			
	ATAIoT7	0.846			
	ATAIoT8	0.882			
	ATAIoT9	0.855			
	ATAIoT10	0.821			
	ATAIoT11	0.850			
	ATAIoT12	0.863			
	ATAIoT13	0.854			
	ATAIoT14	0.825			
Material management	MMA1	0.815	0.944	0.921	0.667
	MMA2	0.788			
	MMA3	0.796			
	MMA4	0.814			
	MMA5	0.835			
	MMA6	0.817			
	MMA7	0.809			
	MMA8	0.816			
	MMA9	0.832			
	MMA10	0.841			
	MMA11	0.823			

Table 5. Correlation of variables–root square of AVE (fornel and larcker result).

Latent variable	ATAGBI	ATAIBS	ATAIoT	CSP	MMA
ATAGBI	**0.875**				
ATAIBS	0.239	**0.836**			
ATAIoT	0.361	0.386	**0.842**		
CSP	0.583	0.623	0.552	**0.825**	
MMA	0.505	0.338	0.342	0.597	**0.814**

Table 6. Heterotrait-monotrait ratio (HTMT).

	ATAGBI	ATAIBS	ATAIoT	CSP	MMA
ATAGBI					
ATAIBS	0.255				
ATAIoT	0.383	0.403			
CSP	0.625	0.649	0.568		
MMA	0.547	0.352	0.357	0.62	

model's endogenous latent variable (CSP) has the value of Q^2 with 0.445, representing an integer value greater than zero. This indicates that the model has 'predictive relevance,' as previously highlighted by Bido and Silva (2019).

Testing the moderating effect (indirect relationship)

Moderation influence exists, in which construct indirectly measured the association between the exogenous and endogenous latent variables significantly (Ramayah et al. 2018). Thus, in this research, MMA is adopted to measure the moderation effects and strength between Automation technology adoptions and construction site performance using Smart PLS 3.7.9. The moderator variable and exogenous latent variables are both assessed reflectively in this research; the product indicator mechanism is

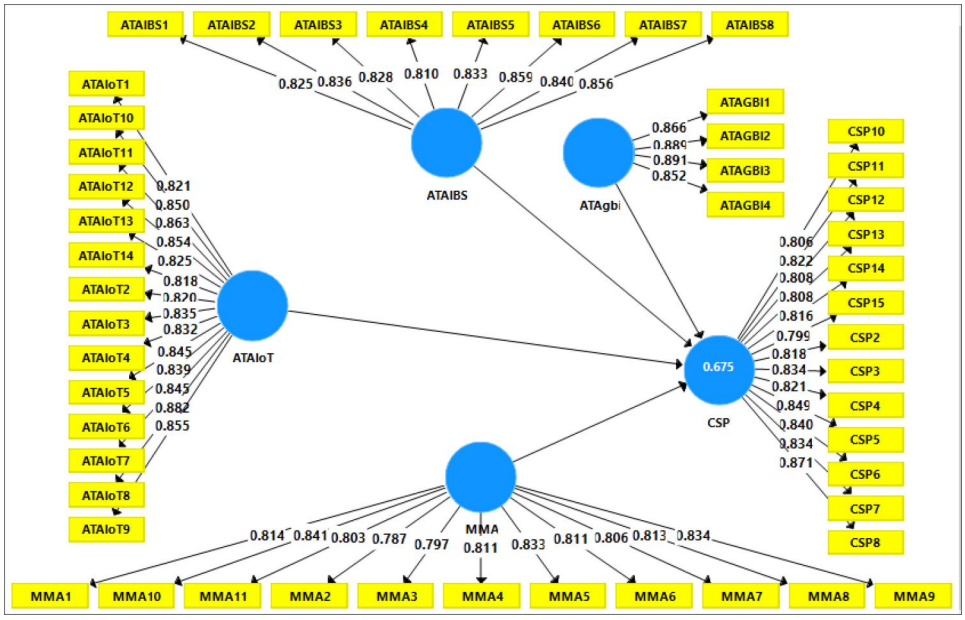


Figure 3. Measurement model.

Table 7. Result of hypothesis testing - direct relationship.

N	Relationship	Std. Beta	Std. Error	*T*- values	*P*-values	Decision
H1	ATAGBI -> CSP	0.289	0.062	4.634	0.000	Supported**
H2	ATAIBS -> CSP	0.388	0.086	4.512	0.000	Supported**
H3	ATAIoT -> CSP	0.213	0.074	2.874	0.004	Supported*
H4	MMA -> CSP	0.248	0.076	3.263	0.001	Supported*

Significant at $*P < 0.05$; $**P < 0.01$.

Table 8. R^2 variance explained in the endogenous construct.

Latent construct	*R*-square	Result
ATAGBI	0.675	High
ATAIBS	0.555	Moderate
ATAIoT	0.641	Moderate
MMA	0.634	Moderate

Table 9. Effect size (f^2) for the latent exogenous construct.

Latent Construct	R^2-Included	R^2- Excluded	F^2	Effect Size
ATAGBI	0.675	0.675	0.181	Moderate
ATAIBS	0.675	0.555	0.373	High
ATAIoT	0.675	0.641	0.107	Small
MMA	0.675	0.634	0.1290	Small

Table 10. Predictive relevance Q^2/cross validity redundancy.

Total	SSO	SSE	1-SSE/SSO
Construction site performance	3341	1854.03	0.445

Table 11. Results of the specific indirect effects.

NO	Relationship	Std. Beta	Std. Error	*T*- value	*P*-value	Decision
H5	ATAGBI->MMA -> CSP	−0.122	0.034	3.609	0000	Supported
H6	ATAIBS->MMA -> CSP	0.100	0.043	2.308	0.021	Supported
H7	ATAIoT->MMA -> CSP	0.068	0.046	1.477	0.140	Not Supported

Significant at $P^*<0.05$; $P^{**}<0.01$.

applicable (Hair and Sarstedt 2019). The specific indirect effect (moderator) findings are presented in Table 11 below.

Table 11 and Figure 4 demonstrate the moderating relationship of MMA between Automation technology adoptions utilizing GBI, IBS, and IoT toward the consequences of construction site performance.

Discussion

This study investigates the impact of automation technology adoption (ATA) on construction site performance (CSP) using the Green Building Index (GBI), Industrialized Building System (IBS), and the Internet of Things (IoT) as key technologies. The objective is to assess the relationship between ATA and CSP while considering the moderating effect of material management adoption (MMA). The hypotheses are tested using data from the survey and supported by relevant literature. The first hypothesis (H1) examines the association between ATA utilizing GBI and CSP. The findings confirmed a significant positive relationship between ATAGBI and CSP ($\beta = 0.289$, $t = 4.634$, $p < 0.000$). This indicates that the adoption of GBI technology for material waste mitigation positively influences the construction site performance. Previous studies have also highlighted the importance of GBI adoption for enhancing construction projects and strengthening market position (Algburi et al. 2016; Rizqa 2016; Aghili 2018; Ali 2018; Vyas et al. 2019; Abkar et al. 2023)).

Green technology adoption, such as GBI, plays a crucial role in preserving the environment and improving the quality of human life for future generations. The second hypothesis (H2) explores the relationship between ATA utilizing IBS and CSP. The findings support H2, indicating a significant positive effect ($\beta = 0.388$, $t = 4.512$, $p < 0.000$). This suggests that IBS technology has a substantial influence on material mitigation approaches and positively impacts construction site performance. These findings align with previous research that emphasizes the benefits of IBS technology in improving construction performance and efficiency (Kamaruddin et al. 2018; Azira et al. 2020; Abkar et al. 2023).

The third hypothesis (H3) investigates the relationship between ATA utilizing IoT and CSP. The finding support H3, indicating a significant positive relationship ($\beta = 0.213$, $t = 2.874$, $p < 0.004$). This underscores the importance of IoT technology in material waste mitigation at the Johor construction sites. Previous studies have shown that IoT technology can enhance material supply chain efficiency and contribute to environmental protection by reducing energy consumption and material wastage (Gbadamosi et al. 2019; Maru and Raval 2020; Perrier et al.

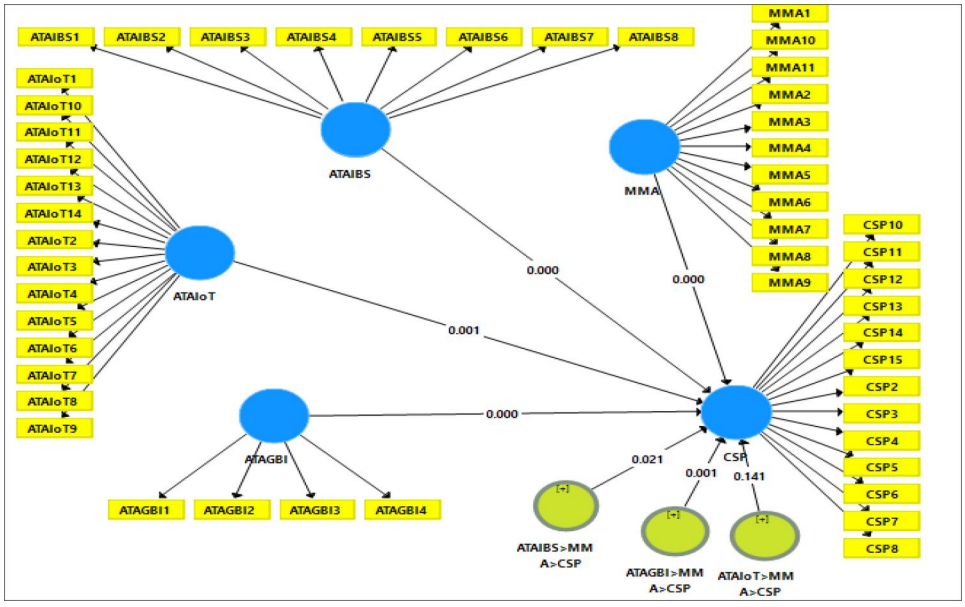


Figure 4. Moderation interaction.

2020; Basheer et al. 2021). The fourth hypothesis (H4) examines the direct association between material management adoption (MMA) and CSP. The findings support H4, indicating a positive influence of MMA on construction site performance ($\beta = 0.248$, $t = 3.263$, $p < 0.001$). Material management practices, including planning, sourcing, and managing materials, have been shown to improve construction efficiency and reduce material wastage, leading to enhanced project profitability (Jereme et al. 2015; Fikri et al. 2020; Umar et al. 2018; Turkyilmaz et al. 2019)

The fifth hypothesis (H5) examines the moderating effect of MMA on the relationship between ATA utilizing GBI and CSP. The findings support H5, indicating a significant negative moderation effect (β=-0.122, $t = 3.609$, $p < 0.000$). This implies that the level of MMA as a moderator strengthens the association between ATAGBI and CSP. The adoption of green building technologies, facilitated by effective material management practices, can mitigate the negative impact of the construction industry on the environment and local inhabitants (Algburi et al. 2016; Latief et al. 2017; Zhineng 2018; Subki and Mahazir 2019; Khoshdelnezamiha et al. 2020).

The sixth hypothesis (H6) examines the moderating effect of MMA on the relationship between ATA utilizing IBS and CSP. The findings support H6, indicating a positive moderation effect ($\beta = 0.100$, $t = 2.308$, $p < 0.021$). Material management practices, when combined with IBS technology, positively influence construction performance. Previous research also emphasizes the importance of material management in enhancing construction site performance (Jereme et al. 2015; Hilmi and Kasim 2017; Núñez-Cacho et al. 2018; Fikri et al. 2020; Turkyilmaz et al. 2019; Kasim et al. 2019; Neyadi 2019; Shi and Xu 2021). However, the seventh hypothesis (H7), which examines the moderating effect of MMA on the relationship between ATA utilizing IoT and CSP, is not supported by the findings. The weak moderation effect ($\beta = 0.068$, $t = 1.477$, $p < 0.140$) suggests that the level of MMA as a moderator does not significantly impact the relationship between ATAIoT and CSP. The implementation of IoT technology in the construction industry is still relatively new and faces challenges (Gamil et al. 2020; Fredriksson et al. 2022). However, previous research has shown the potential of IoT to improve construction management and productivity (Li et al. 2016; Zhong et al. 2017; Xu and Lu 2018; Xu et al. 2018). The evaluation of the overall survey's variables with higher reliability as indicated by Cronbach's Alpha values in Table 2, demonstrates the consistency and accuracy of the questionnaire items, ensuring reliability and validity in measuring the constructs. The study confirmed the reliability and validity of measurement scales, indicating high average variance extracted (AVE) and composite reliability. These validations, spanning diverse contexts and measurements, underscored the scales' reliability and validity. Additionally, the findings significantly contribute to understanding automation technology adoption in the construction industry. They offer actionable insights for the construction sector to prioritize green innovation, improve construction quality, and solidify market position. Moreover, they emphasize the pivotal role of material management in achieving sustainable construction practices and reducing material wastage. Nonetheless, the study's focus remains restricted to examining how automation technology mitigates material waste in Johor's construction, particularly through 3 R, GBI, IBS, and IoT, moderated by material management practices.

Conclusion

In light of the overall research results, this research has attempted to expand the body of knowledge concerning the moderating indication of material management adoption on the relationship between automation technology adoption employing IBS, GBI, and IoT with construction site performance. The findings of this study have been capable of contributing the important theoretical concepts along with certain limitations; it has effectively answered all of the study questions as well as fulfilled all of the stated objectives. The direct hypotheses that investigated the interactions between ATA and CSP were supported in terms of the research model. At the same time, all of the moderating indications were statistically meaningful and confirmed with four hypotheses, the moderation of MMA with ATAIoT on CSP was not supported. Even though many researchers have investigated various underlying aspects of material management on construction sites, the current research has addressed the theoretical gap through the lack of Automation Technology uses

with its indications and absence of material waste mitigation measures at construction sites as a significant influence moderated by MMA indication. This research also provided an empirical evidence for material mitigation measures employing automation technology adoption, along with its three indicators GBI, IBS and IoT. Additionally, it explored the moderating effect of Material Management Adoption (MMA) to examine the relationship between ATA and CSP in Johor's construction sector as part of the Malaysian construction industry. Furthermore, it has highlighted the consequences in terms of environmental and economic aspects for the construction industry, supported by material waste management through the model parameters. Again, based on the theoretical reviews, this research successfully added to the domain of technology innovation for material reduction and construction site enhancement by investigating the influence throughout the construction industry developers, contractors, and experts, as well as the effect of material management applications toward sustainable construction. Aside, the contributions of this research have substantial practical consequences for construction firms and other construction stakeholders. Future research should investigate the effects of automation technology adoption and material waste reduction strategies in construction across diverse regions, shedding light on their influence on industry performance beyond Johor construction firms. Additionally, addressing the unexplained 32.5% variance in construction site performance (CSP) calls for exploring innovative approaches in material management and construction enhancement, including specialized regulatory frameworks, to better comprehend their role in automation technology adoption.

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No potential conflict of interest was reported by the author(s)

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